Genuine Multipartite Entanglement Analysis

Hubbard ring - Convex-roof framework

System parameters

Model	**Hubbard 1D-PBC**
Mode	**uniform**
Sites (L)	**6**
Electrons (Ne)	**6**
Hopping (t)	**1.0**
Interaction (U)	**4.0**
n-body order	**3**
Eigenstate idx	**4 (1-indexed: 5)**

Eigenstate properties

Energy E0	**-2.42291126**
Energy E1	**-2.09275383**
Gap (E1-E0)	**0.330157**
<S^2>	**2.000000**
S	**1.0000**
<Sz>	**0.000000**
Multiplicity	**3.0**
Spin label	**Triplet (S=1)**

Double occupancy

<D> total	**0.471901**
<D> / site	**0.078650**

RDM and GME statistics for this run (3-body clusters)

Hilbert space dim : 2^3 = 8
Number of clusters : 6
Pure clusters : 0
Mixed clusters : 6
Clusters with GME (>0) : 6

Purity gamma = Tr(rho^2) : mean = 0.217185 [min 0.207826 , max 0.231081]
Von Neumann entropy : mean = 1.690155 [min 1.661033 , max 1.706637]
Rank of rho : mean = 8.00 [min 8 , max 8]
C_GME (best estimate) : mean = 0.042363 [min 0.023388 , max 0.069380]

NOTE Analysis performed on eigenstate index 4 (state 5 in 1-indexed convention), not the ground state.

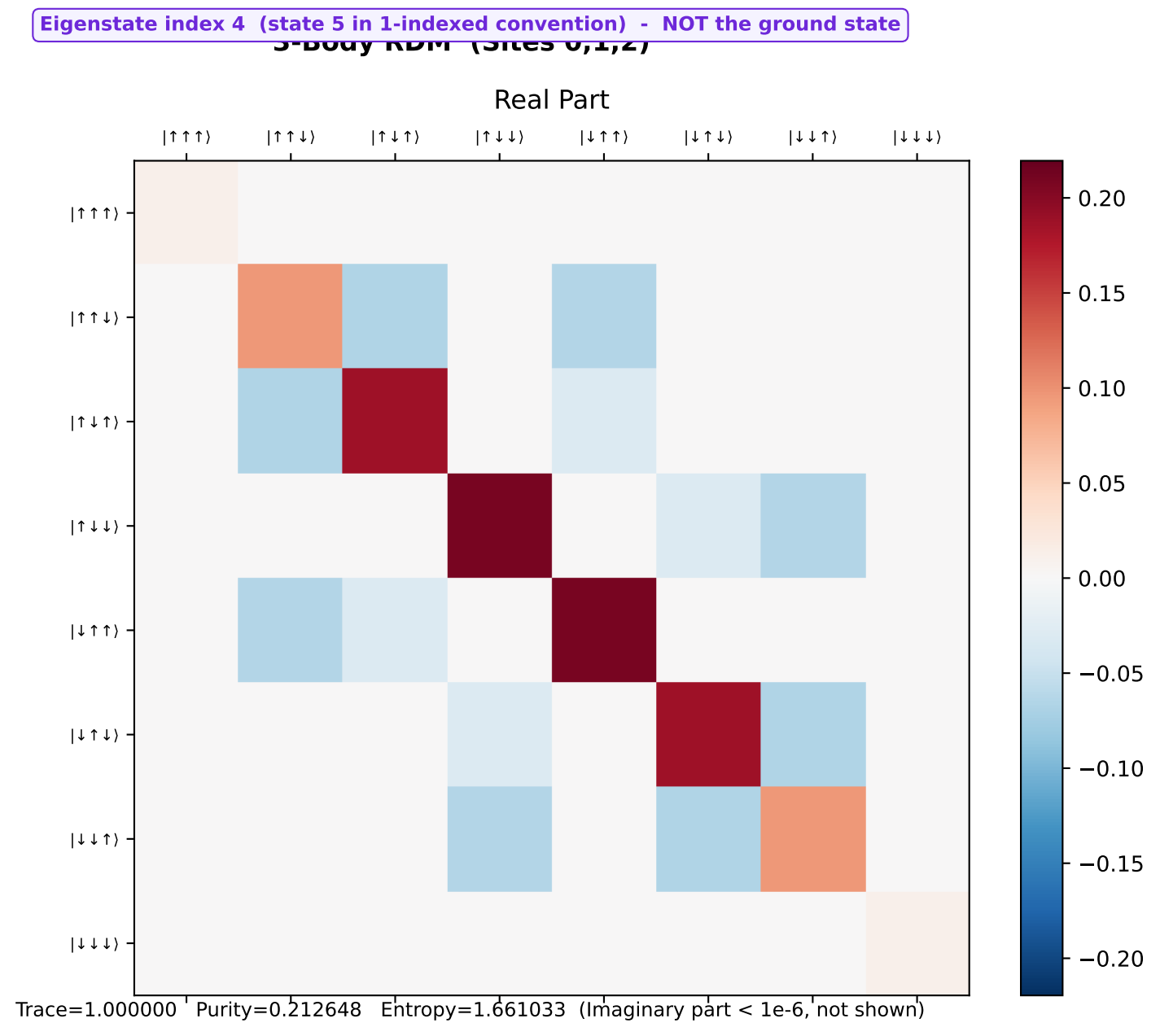
Eigenstate index 4 (state 5 in 1-indexed convention) - NOT the ground state

Summary: All 3-Body RDM Clusters

System: L=6, Ne=6, U=4.0, t=1.0 | E_0=-2.422911

Sites	Purity	C_GME	C_LB	C_cayl	C_expo	C_stiefel	C_UB	Status
(0, 1, 2)	0.212648	–	0.000000	0.000000	0.000000	0.023388	0.217296	MIXED
(1, 2, 3)	0.207826	–	0.000000	0.000000	0.000000	0.046845	0.326665	MIXED
(2, 3, 4)	0.231081	–	0.000000	0.000000	0.000000	0.057968	0.574206	MIXED
(3, 4, 5)	0.212648	–	0.000000	0.000000	0.000000	0.031154	0.217296	MIXED
(4, 5, 0)	0.207826	–	0.000000	0.000000	0.000000	0.025444	0.326712	MIXED
(5, 0, 1)	0.231081	–	0.000000	0.000000	0.000000	0.069380	0.574375	MIXED

Legend:
  Purity = Tr(rho²) [1.0=pure, <1.0=mixed] | C_GME = Genuine multipartite entanglement (pure only)
  C_LB = Lower bound (notes eqs.6-9) | C_cayl = unitary optimization - Cayley Transform |C_expo= unitary optimization - Expoenetial Map | BEST
  C_UB = Upper bound (eigendecomp) | Status: GME =has GME | NO GME=separable | MIXED=mixed state


Eigenstate index 4 (state 5 in 1-indexed convention) - NOT the ground state
3-Body RDM (Sites 0,1,2)
Real Part
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.20
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
−0.20
Trace=1.000000 Purity=0.212648 Entropy=1.661033 (Imaginary part < 1e-6, not shown)

Eigenstate index 4 (state 5 in 1-indexed convention) - NOT the ground state

Analysis Summary - 3-Body RDM (Sites 0,1,2)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.2126482477
  Von Neumann entropy  : 1.6610328136
  Eigenvalue min       : 1.037306e-02
  Eigenvalue max       : 2.382841e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 0.21264825

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

STATUS: MIXED STATE
------------------------------------

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

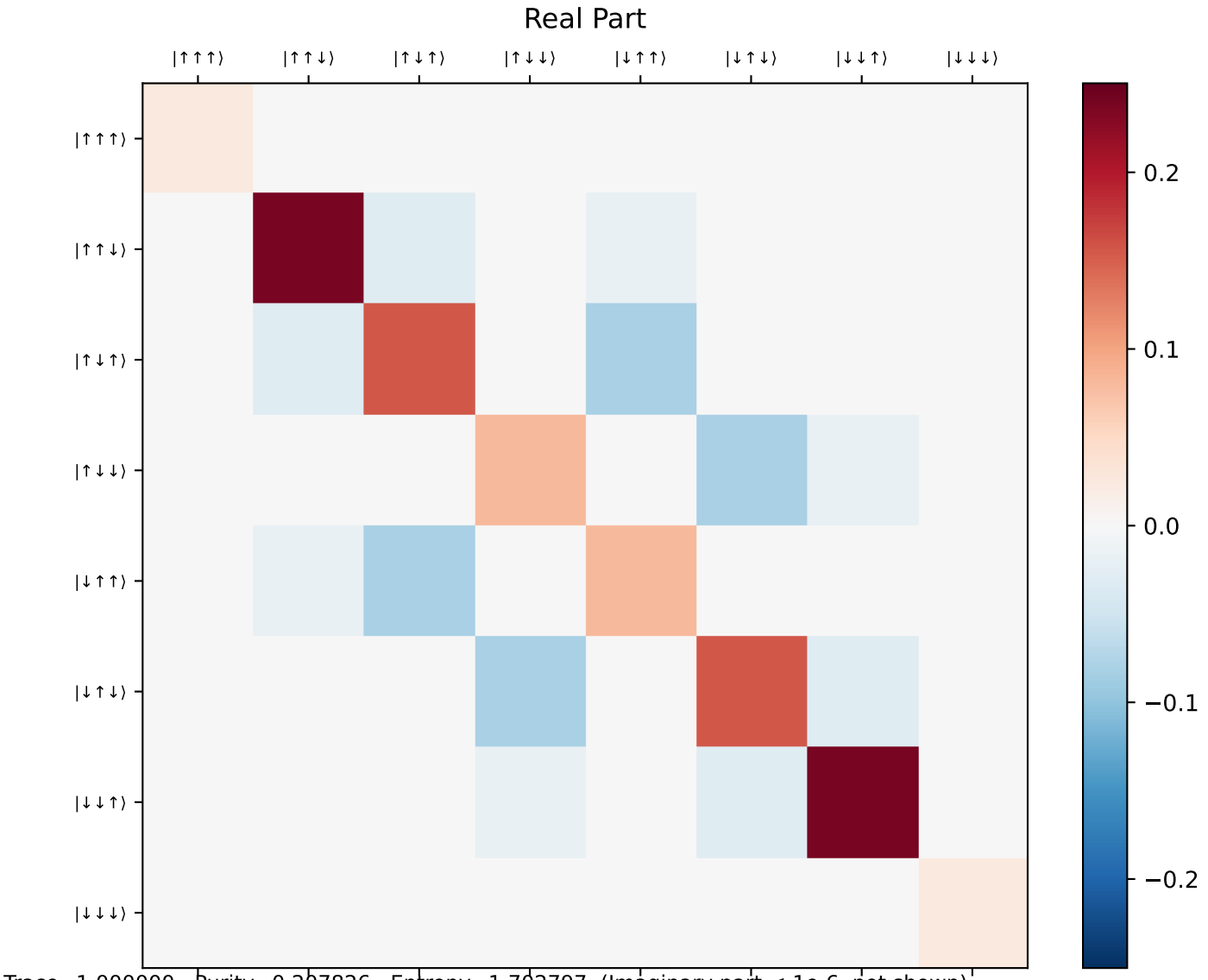
```
Convex Roof Analysis
====================================

  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = 0.02338797   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.21729597

  Status: nONE
```

Eigenstate index 4 (state 5 in 1-indexed convention) - NOT the ground state

3-Body RDM (Sites 1,2,3)



Trace=1.000000 Purity=0.207826 Entropy=1.702797 (Imaginary part < 1e-6, not shown)

Eigenstate index 4 (state 5 in 1-indexed convention) - NOT the ground state

Analysis Summary - 3-Body RDM (Sites 1,2,3)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.2078258125
  Von Neumann entropy  : 1.7027967845
  Eigenvalue min       : 2.413820e-02
  Eigenvalue max       : 2.484999e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 0.20782581

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

STATUS: MIXED STATE
------------------------------------

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

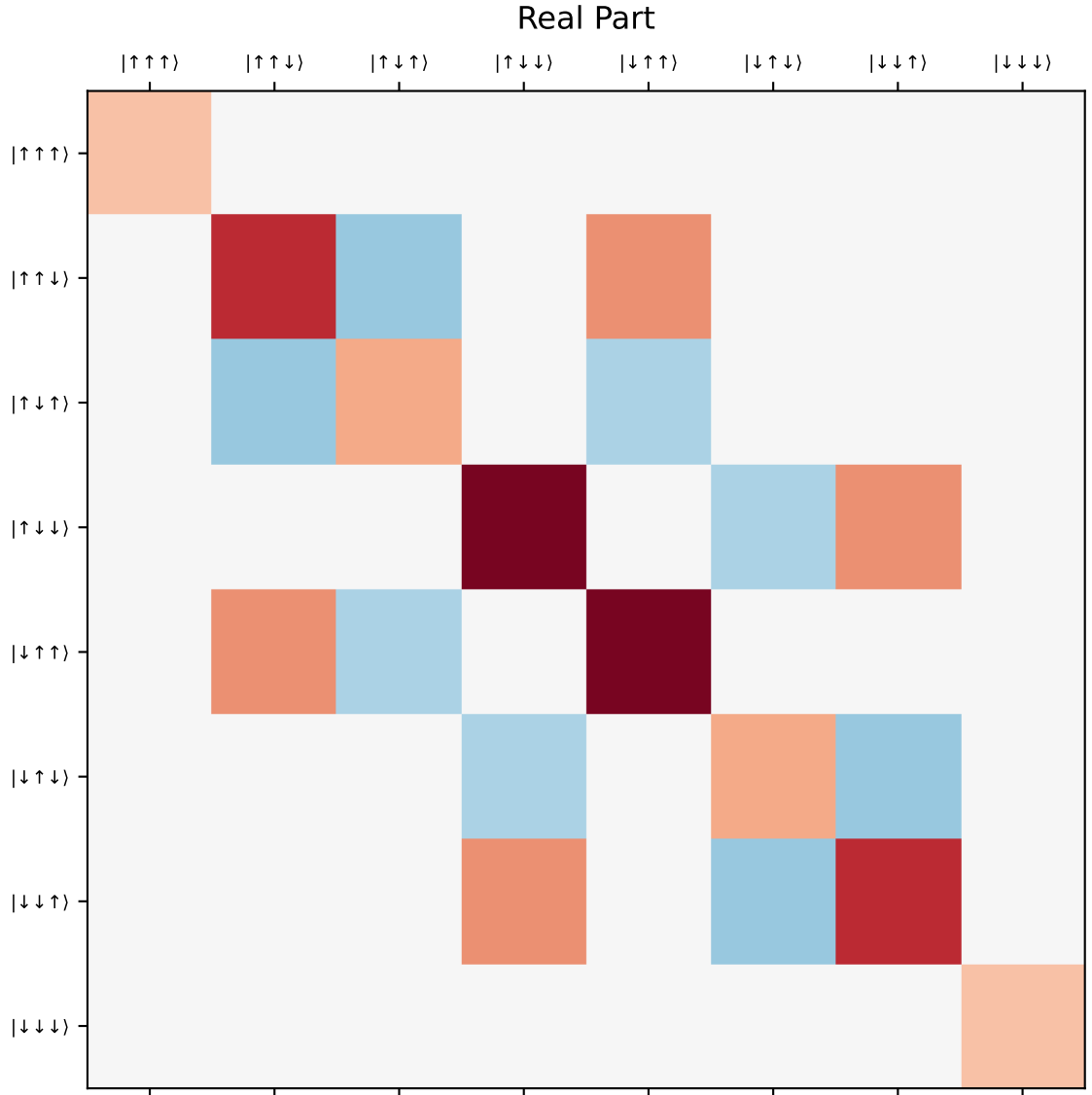
```
Convex Roof Analysis
====================================

  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = 0.04684542   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.32666498

  Status: nONE
```

Eigenstate index 4 (state 5 in 1-indexed convention) - NOT the ground state

3-Body RDM (Sites 2,3,4)



Trace=1.000000 Purity=0.231081 Entropy=1.706637 (Imaginary part < 1e-6, not shown)

Eigenstate index 4 (state 5 in 1-indexed convention) - NOT the ground state

Analysis Summary - 3-Body RDM (Sites 2,3,4)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.2310806182
  Von Neumann entropy  : 1.7066367758
  Eigenvalue min       : 2.863319e-02
  Eigenvalue max       : 3.211104e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 0.23108062

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

STATUS: MIXED STATE
------------------------------------

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

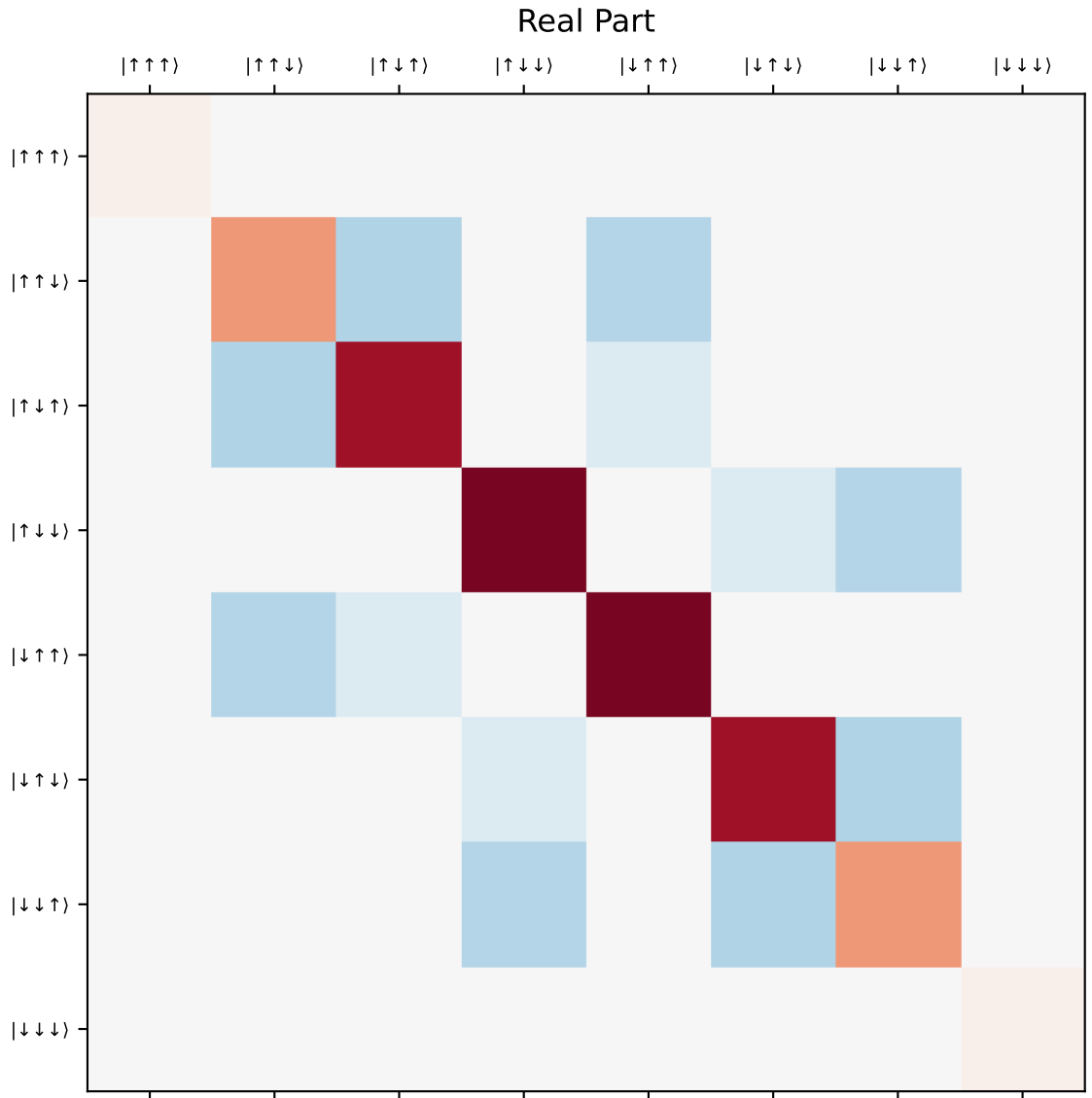
```
Convex Roof Analysis
====================================

  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = 0.05796848   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.57420643

  Status: nONE
```

Eigenstate index 4 (state 5 in 1-indexed convention) - NOT the ground state

3-Body RDM (Sites 3,4,5)



Trace=1.000000 Purity=0.212648 Entropy=1.661033 (Imaginary part < 1e-6, not shown)

Eigenstate index 4 (state 5 in 1-indexed convention) - NOT the ground state

Analysis Summary - 3-Body RDM (Sites 3,4,5)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.2126482477
  Von Neumann entropy  : 1.6610328136
  Eigenvalue min       : 1.037306e-02
  Eigenvalue max       : 2.382841e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 0.21264825

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

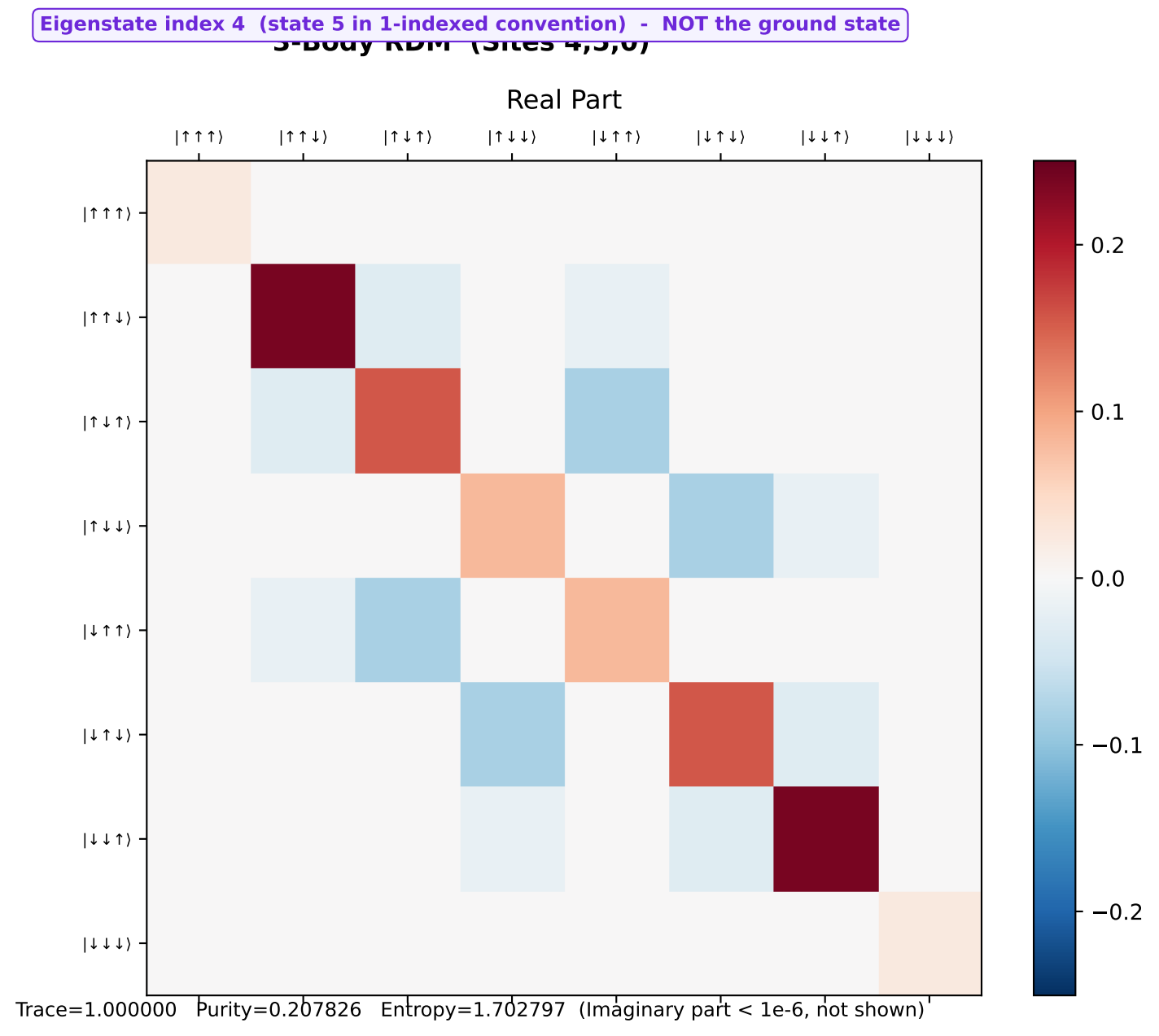
STATUS: MIXED STATE
------------------------------------

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

```
Convex Roof Analysis
====================================

  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = 0.03115369   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.21729596

  Status: nONE
```


Eigenstate index 4 (state 5 in 1-indexed convention) - NOT the ground state
3-Body RDM (Sites 4,3,0)
Real Part
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=0.207826 Entropy=1.702797 (Imaginary part < 1e-6, not shown)

Eigenstate index 4 (state 5 in 1-indexed convention) - NOT the ground state

Analysis Summary - 3-Body RDM (Sites 4,5,0)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.2078258125
  Von Neumann entropy  : 1.7027967845
  Eigenvalue min       : 2.413820e-02
  Eigenvalue max       : 2.484999e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 0.20782581

  A            | B            | Tr(rho_A^2)
  ____________ | ____________ | ___________

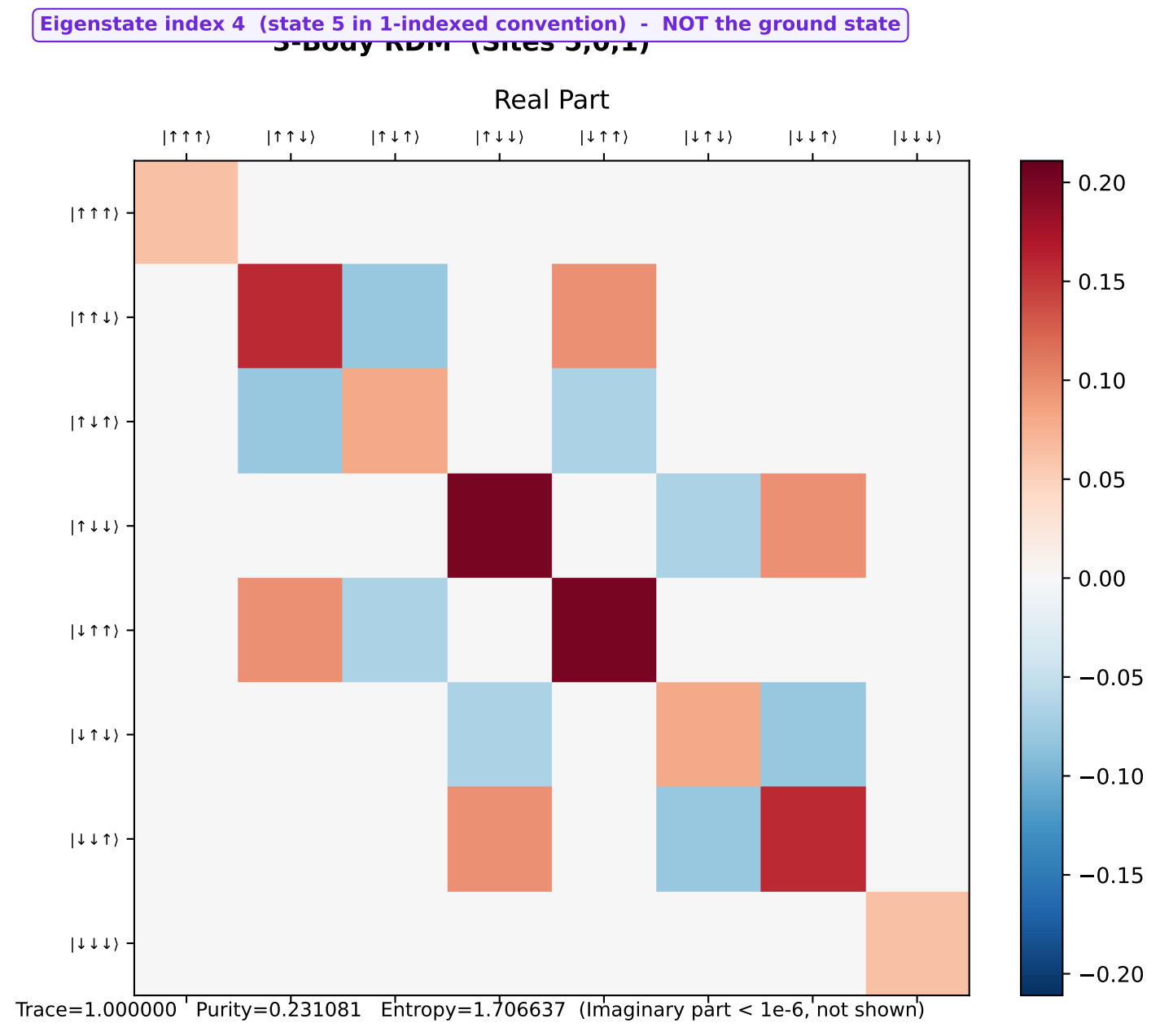
STATUS: MIXED STATE
____________________________________

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

```
Convex Roof Analysis
====================================

  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = 0.02544415   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.32671181

  Status: nONE
```


Eigenstate index 4 (state 5 in 1-indexed convention) - NOT the ground state
Real Part
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.20
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
−0.20
Trace=1.000000 Purity=0.231081 Entropy=1.706637 (Imaginary part < 1e-6, not shown)

Eigenstate index 4 (state 5 in 1-indexed convention) - NOT the ground state

Analysis Summary - 3-Body RDM (Sites 5,0,1)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.2310806182
  Von Neumann entropy  : 1.7066367758
  Eigenvalue min       : 2.863319e-02
  Eigenvalue max       : 3.211104e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 0.23108062

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

STATUS: MIXED STATE
------------------------------------

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

```
Convex Roof Analysis
====================================

  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = 0.06937952   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.57437496

  Status: nONE
```